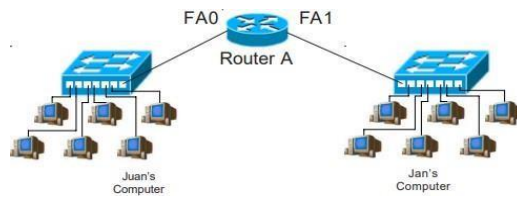


1. What is an Access Control List (ACL) in networking?
 - It is a set of rules that control network traffic.
 - It defines whether to permit or deny packets based on source and destination IP addresses, protocol types, and other packet attributes.
 - ACLs are used to enhance security by filtering traffic in a network.
2. What are the common types of ACL? There are two primary types of ACLs.
 - (1) Standard ACLs: Filter traffic based on the source IP address only.
 - (2) Extended ACLs: Filter traffic based on source/destination IP addresses, protocols and port numbers.
3. Where should you apply a standard ACL in a network?
 - Standard ACLs should be applied as close to the destination as possible because they filter traffic based only on the source IP address.
 - Applying them too close to the source might unintentionally block legitimate traffic.
4. Where should you apply an extended ACL in a network?
 - Extended ACLs should be applied as close to the source as possible.
 - This minimizes unnecessary traffic in the network by filtering out unwanted traffic before it traverses the network.
5. What is the difference between a numbered ACL and a named ACL?
 - Numbered ACL: An ACL identified by a number (1-99 for standard, 100-199 for extended).
 - Named ACL: An ACL identified by a descriptive name. Named ACLs provide more flexibility as they allow adding or removing specific rules without removing the entire ACL.
6. What is the implicit "deny all" rule in ACLs?
 - There is an implicit "deny all" rule at the end of every ACL.
 - This means that if a packet does not match any of the permit rules in the ACL, it will be denied by default.
7. How does an ACL process traffic?
 - It is in a sequential manner, from top to bottom.
 - When a packet matches a rule, the ACL stops processing further rules.
 - If the packet does not match any rule, the implicit "deny all" rule applies.

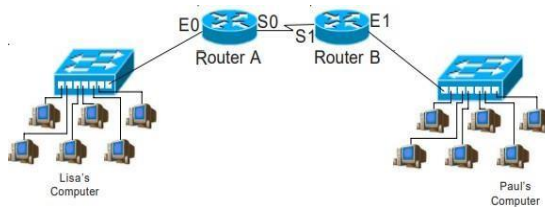
8. What is the difference between "inbound" and "outbound" ACLs?
- Inbound ACL: Filters traffic entering an interface before it is routed.
 - Outbound ACL: Filters traffic after it has been routed, as it exits the interface.
9. How do ACLs enhance network security?
- ACLs enhance network security by restricting unauthorized access, controlling the flow of traffic, and preventing specific types of traffic from entering or leaving a network.
 - They can be used to prevent attacks, limit access to critical resources, and enforce network policies.
10. What is the impact of incorrectly configured ACLs?
- Incorrectly configured ACLs can block legitimate traffic, leading to network outages or degraded performance.
 - They can also fail to block malicious traffic, leaving the network vulnerable to attacks.
11. Write the Standard ACL for Basic Traffic Filtering. Block traffic from a specific host. Create a standard ACL (numbered 1-99) to deny traffic from host 192.168.1.10. Apply the ACL to inbound traffic on interface GigabitEthernet0/0.
- access-list 10 deny host 192.168.1.10
 - access-list 10 permit any
 - interface GigabitEthernet0/0
 - ip access-group 10 in
12. Write the extended ACL for Specific Services. Allow HTTP/HTTPS traffic to a web server. Create an extended ACL (numbered 100-199) to permit HTTP/HTTPS to 10.0.0.5. Apply the ACL to outbound traffic on interface GigabitEthernet0/0.
- access-list 110 permit tcp any host 10.0.0.5 eq 80
 - access-list 110 permit tcp any host 10.0.0.5 eq 443
 - access-list 110 deny ip any any
 - interface GigabitEthernet0/0
 - ip access-group 110 out

13. In order to permit packets from Juan's computer to arrive at Jan's computer you would place the standard access list at router interface .



Answer: FA1

14. Lisa has been sending unnecessary information to Paul.



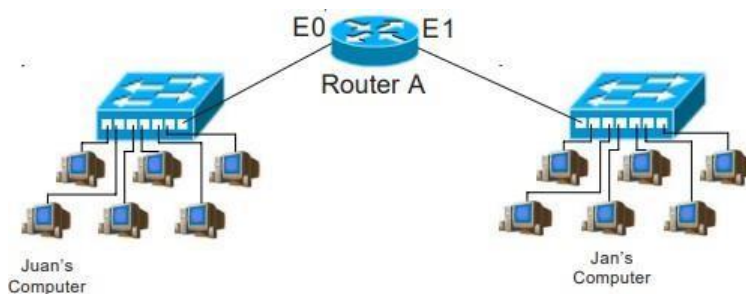
Where would you place the standard ACL to deny all traffic from Lisa to Paul?

Answer: Router B, E1

Where would you place the standard ACL to deny traffic from Paul to Lisa?

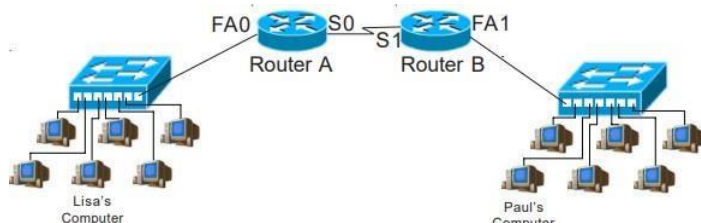
Answer: Router A, E0

15. In order to permit packets from Juan's computer to arrive at Jan's computer you would place the extended access list at router interface .



▪ Answer: E0

16. Lisa has been sending unnecessary information to Paul. Where would you place the extended ACL to deny all traffic from Lisa to Paul?

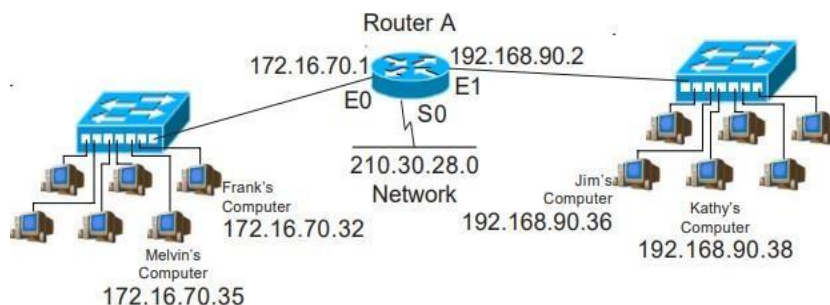


- Answer: Router A, FA0

Where would you place the extended ACL to deny traffic from Paul to Lisa?

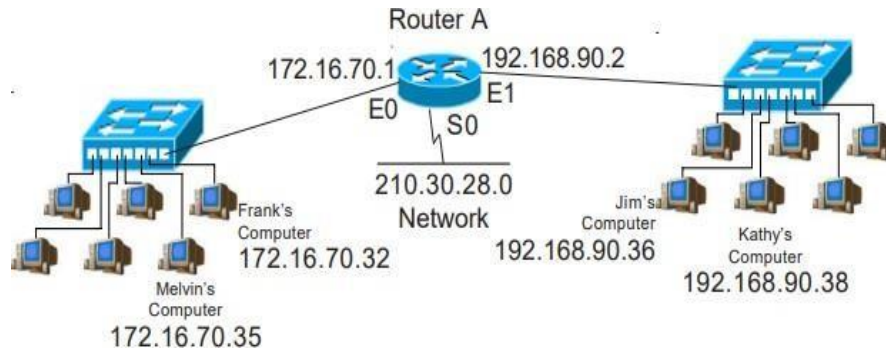
- Answer: Router B, FA1

17. Write a named standard access list called “GEO” on Router A, interface E1 to block Melvin’s computer from sending information to Kathy’s computer; but will allow all other traffic. Place the access list at

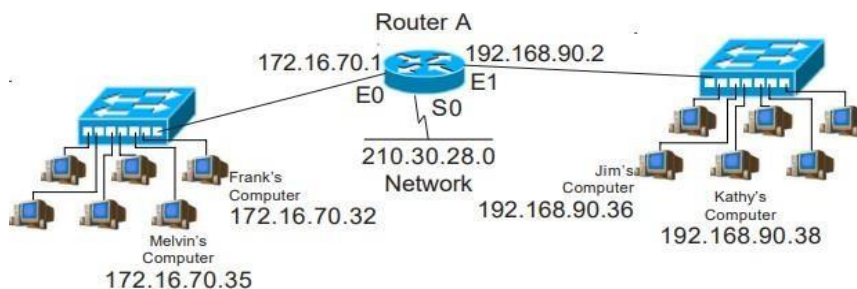


- ip access-list standard GEO
- deny host 172.16.70.35
- access-list permit any
- interface E1
- ip access-group GEO out

18. Write a standard access list to block Melvin's computer from sending information to Kathy's computer; but will allow all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written. Place the access list at:



- access-list 10 deny host 172.16.70.35
 - access-list 10 permit any
 - interface E1
 - ip access-group 10 out
19. Write a standard access list to block Jim's computer from sending information to Frank's computer; but will allow all other traffic from the 192.168.90.0/24 network. Permit all traffic from the 210.30.28.0/24 network to reach the 172.16.70.0/24 network. Deny all other traffic. Keep in mind that there may be multiple ways many of the individual statements in an ACL can be written. Place the access list at:



- access-list 25 deny 192.168.90.36
- access-list 25 permit 192.168.90.0 0.0.0.255
- access-list 25 permit 210.30.28.0 0.0.0.255
- interface E0
- ip access-group 25 out

20. Create an ACL that allows HTTP (TCP port 80) traffic from the source network 192.168.11.0/24 to the destination network 192.168.20.0/24. Deny all other traffic from 192.168.11.0/24 to 192.168.20.0/24. Apply the ACL to the correct interface.

- access-list 100 permit tcp 192.168.11.0 0.0.0.255 192.168.20.0 0.0.0.255 eq 80
- access-list 100 deny ip 192.168.11.0 0.0.0.255 192.168.20.0 0.0.0.255
- access-list 100 permit ip any any

21. Create an extended ACL to allow only ICMP and HTTP traffic from 192.168.20.0/24 to 192.168.40.0/24. Deny all other traffic.

- access-list 101 permit icmp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
- access-list 101 permit tcp 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255 eq 80
- access-list 101 deny ip 192.168.20.0 0.0.0.255 192.168.40.0 0.0.0.255
- access-list 101 permit ip any any

22. Create an extended ACL that denies all traffic from the source IP address 192.168.50.10 to any destination, but allows all other traffic.

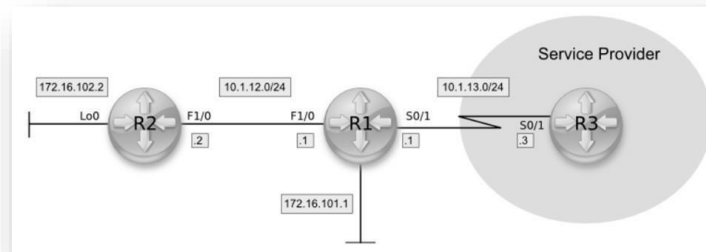
- access-list 102 deny ip host 192.168.50.10 any
- access-list 102 permit ip any any

23. Create an ACL that allows DNS (port 53), HTTP (port 80), and HTTPS (port 443) traffic from the network 192.168.100.0/24 to any destination but denies all other traffic.

- access-list 105 permit udp 192.168.100.0 0.0.0.255 any eq 53
- access-list 105 permit tcp 192.168.100.0 0.0.0.255 any eq 80
- access-list 105 permit tcp 192.168.100.0 0.0.0.255 any eq 443
- access-list 105 deny ip 192.168.100.0 0.0.0.255 any
- access-list 105 permit ip any any

24. You are in charge of R1 and R2 routers. R3 belongs to your Service Provider network and simulates Internet.

- (i) Configure an Extended access-list disabling anyone TELNET to R1 and all devices behind it (R2) if the traffic is originated from Internet (here: SP). All other traffic should be permitted.



- access-list 100 deny tcp any any eq telnet
- access-list 100 permit ip any any
- interface s0/1
- ip access-group 100 in

- (ii) On R1 remove the previous ACL and configure a new one allowing only HTTP access to 172.16.102.0/24 if the traffic is originated from Internet (here: SP). All other traffic should be discarded.

- no access-list 100
- interface s0/1
- no ip access-group 100 in
- access-list 101 permit tcp any host 172.16.102.2 eq www
- interface s0/1
- ip access-group 101 in

25. Write the Wildcard Mask for specific service. Allow only the 192.168.4.0/24 subnet. Write a standard ACL and use a wildcard mask to match the subnet:

- access-list 20 permit 192.168.4.0 0.0.0.255